AI TUTOR PROJECT

A RAG-Based Knowledge Engine for Personalized Education

The RAG Advantage

Retrieval-Augmented Generation bridges the gap between static model weights and dynamic, domain-specific data.

Big Picture Vision

Local Intelligence

An AI Tutor that resides locally, processing course videos and storing conceptual meaning as high-dimensional vectors for instant, private retrieval.

Contextual Retrieval

Moving beyond keywords to semantic intent. Understands that "how to loop" relates to "iteration" segments even without exact phrasing.

Project Roadmap

Ingestion

Video to Audio using FFmpeg

Transcription

Whisper Speech-to-Text

Encoding

BGE-M3 Chunking & Embeddings

Storage

NumPy Array Vector Store

Response

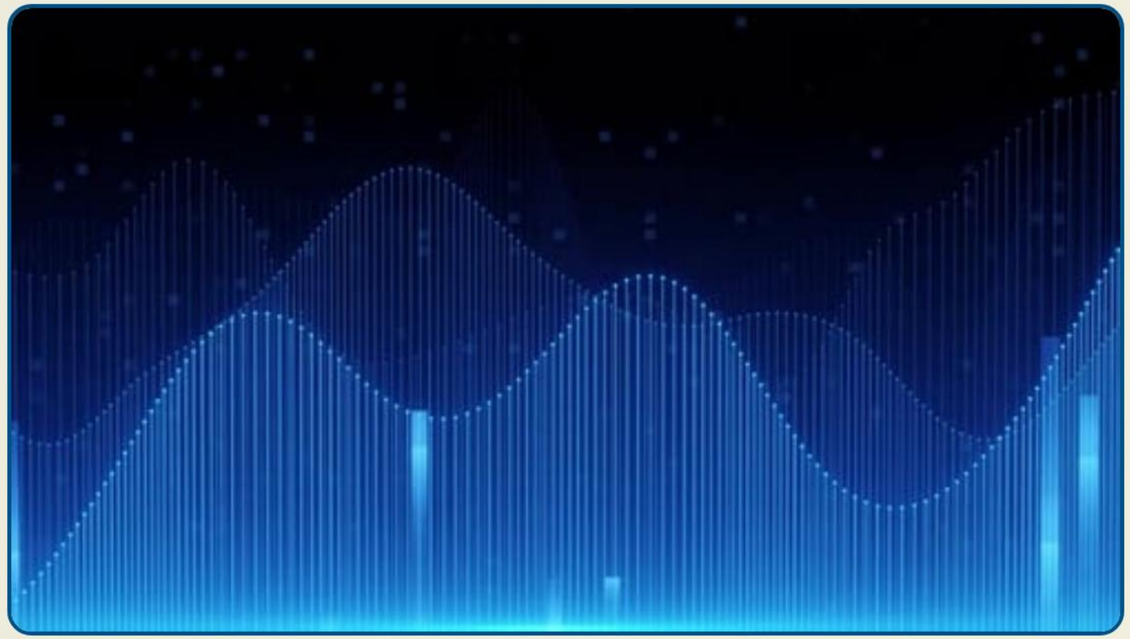
Llama 3.2 AI Brain

The "Ears": Transcription

OpenAI Whisper Integration

Whisper processes raw video audio to generate precise, timestamped text transcripts. This transforms unstructured media into a searchable text dataset.

Analogy: *Recording a lecture and having a perfect, verbatim stenographer at your side.*



The Encoding Pipeline

Chunking

Breaking long transcripts into 200-400 word segments. Smaller "flashcards" allow for precise relevance during retrieval.

Embeddings

Using BGE-M3 to convert text chunks into numeric vectors. Encodes semantic meaning into mathematical GPS coordinates.

Vector Store

Storing embeddings locally via **NumPy and Joblib**. High-performance matrix storage becomes the system's "Local Library."

Semantic Precision

1024

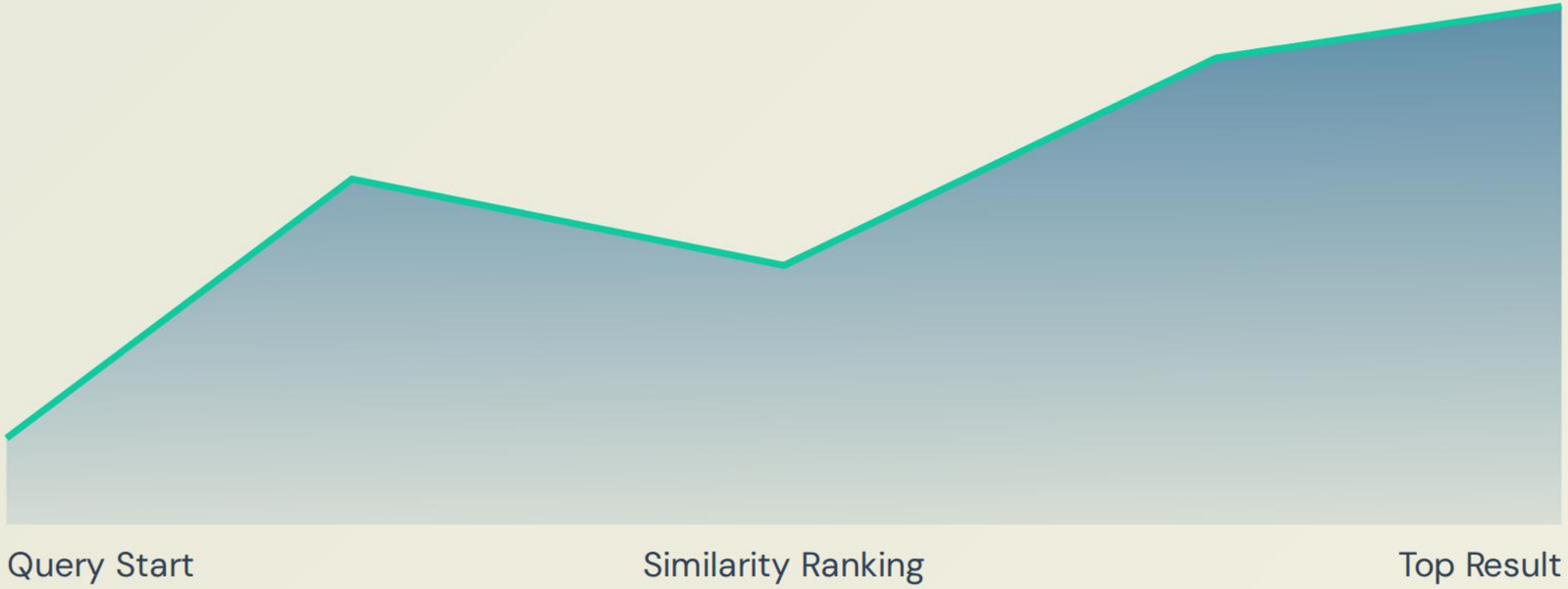
Embedding Dimensions

The Math of Meaning

Every chunk of text is mapped into a high-dimensional space. By calculating the **Cosine Similarity**, we find the segments mathematically closest to a student's question.

$$S(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

Retrieval Confidence Scores



The system retrieves context with over 90% semantic match for domain-specific Python queries.

The Brain: Llama 3.2

Reasoning Over Context

Llama 3.2 acts as the expert explainer. It reads the specific chunks provided by the NumPy vector search and synthesizes a human-like answer.

- **No Hallucinations:** Grounded in data.
- **Timestamp Support:** Exact references.
- **Local Privacy:** Data never leaves.

System Mapping Analogies

System Component	Role in Project	Real World Analogy
Whisper	Ears / Transcription	Lecture Recording & Notes
BGE-M3	Encoding / Meaning	The Librarian's Index
NumPy / Joblib	Library / Storage	Shelf Full of Flashcards
Llama 3.2	Brain / Reasoning	A Smart Teaching Assistant

The Single Question Lifecycle

Input

User submits Python question.

Embed

BGE-M3 creates query vector.

Retrieve

Top chunks from NumPy.

Generate

Llama 3.2 writes answer.

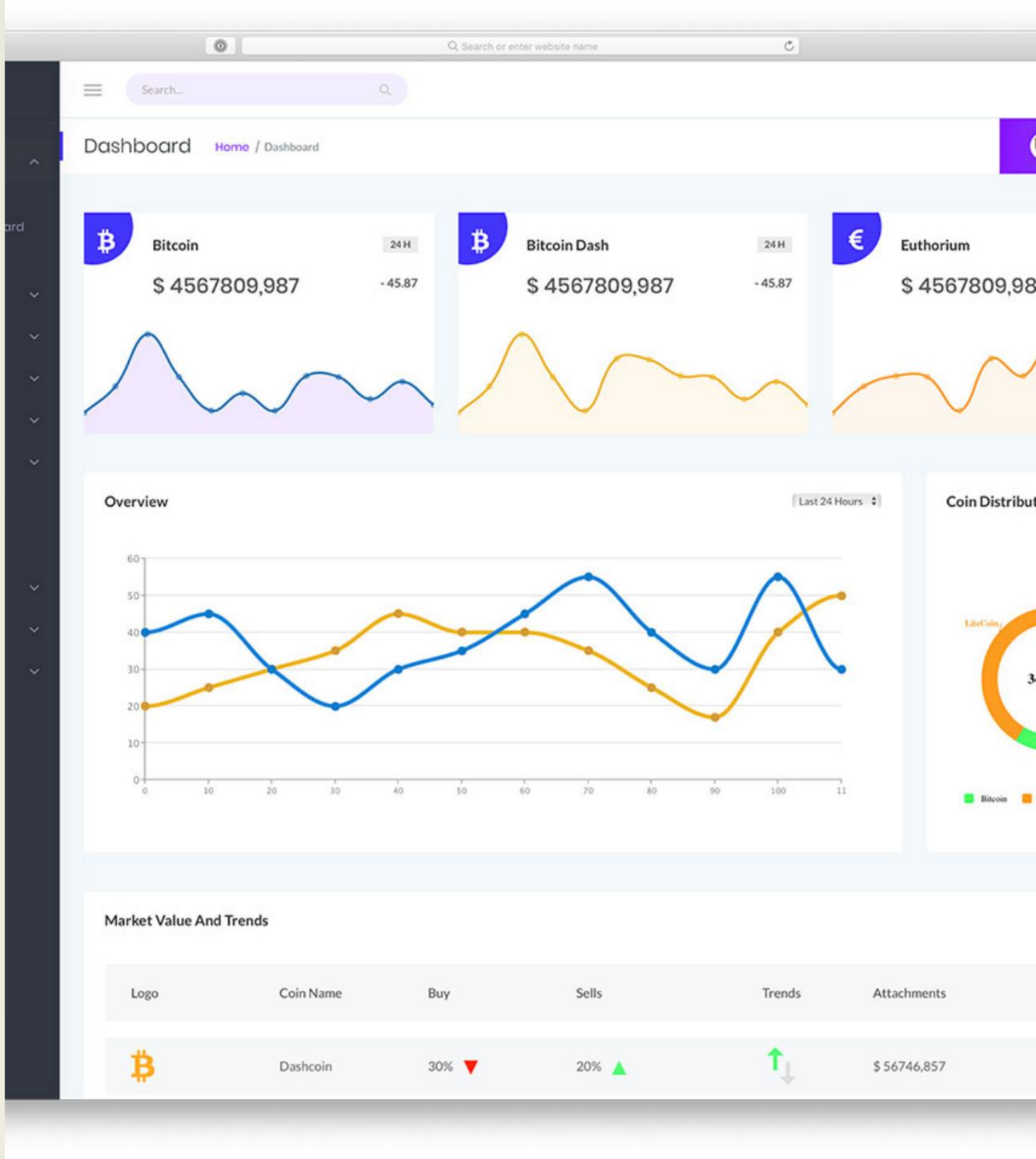
600 × 400

The Interface: Flask & UI

The system's **"Nervous System"** is built on Flask, serving as the API bridge between the NumPy-based logic and the front-end.

HTML and CSS provide the **"Face"** of the project—a clean, interactive classroom board where students query the video library seamlessly.

> … …



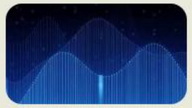
Q&A

Exploring the Future of RAG-Based Education

System v1.0 | Local Llama 3.2 | Ollama Backend

Optimized for localized knowledge with NumPy matrix storage.

Image Sources



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